

# Ramaakshay Mallireddy

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## OBJECTIVE

Aerospace Engineering student seeking an AI/ML Engineer or Software Engineer role, with hands-on experience building ML/LLM systems, full-stack applications, and cloud infrastructure. Experienced designing and evaluating RL and IL systems end-to-end, from training pipelines through benchmarked deployment.

## EDUCATION

### University of Texas at Austin

*B.S. Aerospace Engineering | Minors: Computer Science, Robotics*

Austin, TX

*Expected Graduation: Spring 2028*

### Relevant Coursework:

- Data Structures & Algorithms
- Vector Calculus & Linear Algebra
- AI Design and Development
- Machine Learning in Robotics

## EXPERIENCE

### AI Forward Deployed/Software Engineer Intern

May 2026 – August 2026

*IBM*

*Austin, TX*

- Designed end-to-end AI solutions on IBM's Client Engineering team, translating business requirements into cloud-native proof-of-concept deliverables
- Provisioned cloud infrastructure with Python, Terraform, and Ansible, and deployed containerized services (Docker/Kubernetes/ECS/Lambda, serverless) across Red Hat OpenShift, IBM Cloud, GCP, and AWS; wrote CI/CD pipelines (Jenkins, Bash) for repeatable deployments
- **IBM WorkTrace** — Full-stack RAG platform for time reporting and team activity tracking, independently built and adopted across intern and early-hire programs
  - Built semantic search and multi-provider LLM inference (watsonx.ai, Groq, Ollama) with Python, FastAPI, React, Node, and pgvector; used GitLab CI/CD for deployment
  - Built an agentic pipeline to parse transcripts, classify tasks, and route structured time entries and data models into a PostgreSQL data warehouse, with automated ETL testing and refactoring
  - Presented progress and design decisions to engineering leadership, gathering feedback to prioritize the roadmap and drive adoption across intern and early-hire programs in the US
- **IBM Bobathon — 1st Place** (1,000+ participants): AI-powered IBM Concert workflow monitor on OpenShift that classifies failures, auto-remediates incidents, and alerts Slack via a live dashboard; presented to VPs and Neel Sundaresan, creator of Bob and watsonx Orchestrate

### Texas Spacecraft Laboratory (Electrical Power Systems)

August 2025 – Present

*University of Texas at Austin*

*Austin, TX*

- Designed EGSE PCB for spacecraft power system testing; modeled enclosure/mounting geometry in SolidWorks/CAD for design for manufacturability
- Selected and tested 15+ power components (batteries, solar panels, converters) using multimeters, oscilloscopes, and power supplies; troubleshooted hardware faults and signal integrity issues via datasheet interpretation and driver writing
- Ran Python-based orbital energy simulations across mission scenarios, validating power consumption models against bench data through data collection, analysis, and reporting

## PROJECTS & RESEARCH

### IL & RL Blended Recovery Architecture for Robotic Manipulation | *PyTorch, TensorFlow*

January 2026 – Present

- Built a dual-policy robotic control framework combining a behavior cloning (IL) policy with a deep RL recovery policy (actor-critic) to handle out-of-distribution states, using state estimates from a TensorFlow vision module
- Implemented a safety-constrained action-projection shielding layer to bound control inputs and prevent collisions; evaluated on collision rate, recovery latency, and task success
- Ran automated testing and iterative refactoring passes across the development life cycle to keep the training and evaluation codebase maintainable as the architecture evolved

### RL Drone Swarm Task Allocation via Decentralized Auction | *PyTorch, PPO, Python*

May 2026 – Present

- Designing a decentralized auction system where each drone runs a fixed RL policy per task type while a learned PPO bidding policy handles real-time task reassignment as missions shift
- Modeled the airframe in SolidWorks/CAD and ran CFD for drag/lift; benchmarked over 1,320 episodes against CBBA/Greedy/Oracle baselines, reaching 100% task completion with near-zero variance on 5 of 6 disruption scenarios and up to 260x lower reward variance than Greedy under stochastic conditions

## TECHNICAL SKILLS

**AI/ML Tools:** PyTorch, TensorFlow, watsonx.ai, LangChain, Hugging Face, OpenAI API, scikit-learn, FAISS, OpenCV

**Cloud & Systems:** AWS, GCP, IBM Cloud, Docker, Kubernetes/OpenShift, Terraform, Jenkins/CI/CD, Linux

**Languages:** Python, Go, Rust, Java, C/C++, C#, SQL, JavaScript/TypeScript

**Data & Web:** PostgreSQL, Snowflake, React, Next.js, REST APIs, pandas, NumPy

**Other:** SolidWorks/CAD, CFD, Bash, Git